Timing validation: five scenarios

Objective

Test event generation, reception and timestamp agreement on the CXFEL digital timing rack. Test one event, a two-event gap, multiple delays, repeated Normal-mode execution, and camera response to an event-rate change.

The report contains the recorded results and graphics. It can be read without access to the test rack or its control interface.

Setup

All test chassis ran firmware and IOC release 20261017. The reference clock was 125 MHz. Receiver timestamps use 8 ns ticks. Nanosecond units do not establish 1 ns accuracy or absolute UTC accuracy.

Equipment

Function	IP address	PV prefix	Configuration
Generator / local receiver	10.139.5.196	EVT:DEV2:20:	EVG/EVR role; sequencer bank 1
Compared receiver A	10.139.5.195	EVT:DEV2:10:	Test-event logging
Output receiver	10.139.5.197	EVT:DEV2:30:	Output 1 selects event 101
Compared receiver B	10.139.5.198	EVT:DEV2:40:	Test-event logging
CAM2	192.168.2.1	CAM:DEV2:20:	External Line1 rising-edge FrameStart; 1 ms exposure; 1200 × 900 Mono12
DG645 units	10.139.4.87 / 10.139.4.88	DGEN:DEV2:20: / DGEN:DEV2:30:	LED breadboard attached; connected unit/output unrecorded
IOC / analysis host	10.139.4.59	—	Camera network interface 192.168.2.100

The exact fiber ports and camera-trigger cable endpoints were not recorded. Output 1 on 10.139.5.197 is the configured event-101 pulse output. Its cable destination remains unverified. The records do not establish physical cable skew.

Procedure

1. Read each receiver status. Require link-up and timestamp-valid bits. Require the FIFO overflow bit to be clear. Save the original controls and event maps.
2. Check that test codes 42 and 43 are unused. Enable them for logging on 10.139.5.195 and 10.139.5.198. Keep their pulse-output action bits clear.
3. For scenarios 1–3, load the listed bank-1 table in Single mode. Arm and issue one software trigger. Code 255 ends the table and is not transmitted.
4. For scenario 4, load the two-event table in Normal mode. Keep timer and input trigger masks zero. Arm once. Issue 20 software triggers, nominally 100 ms apart. After each trigger, verify one counter increment and idle/armed state.

5. The bank counter is four bits and wraps modulo 16. Verify every increment; a final raw counter value alone cannot count a 20-cycle trial.
6. For scenario 5, set event 101 to 1 Hz. Keep output 1 on 10.139.5.197 at 0 ns delay and 100,000 ns width. Keep camera acquisition and DG645 settings unchanged.
7. Record full receiver logs and camera metadata. Save lossless images for the camera test. Use the full recording for analysis, not a rolling display buffer.
8. Match test codes, counts and order on both compared receivers. For scenario 4, calculate each gap from that cycle's first event. Subtract A timestamps from B.
9. Disarm and wait for idle/disarmed readbacks. Restore the starting settings. Compare fresh controls and full event maps with the saved originals.

Test definitions

Scenario	Event table or setting	Capture
1 · Single event	42 at 0 ns; stop at 1,000 ns	One execution; 10 s recording
2 · Two-event gap	42 at 0 ns; 43 at 1,000 ns; stop at 2,000 ns	One execution; 60 s recording
3 · Multiple delays	42 at 0 ns; 43 at 1,000 ns; 42 at 10,000 ns; stop at 11,000 ns	One execution; 10 s recording
4 · Repeated sequence	42 at 0 ns; 43 at 1,000 ns; stop at 2,000 ns	20 Normal-mode executions; 10 s recording
5 · LED / camera	Event 101 at 1 Hz; output width 100,000 ns	60 s; full images saved

Event times are offsets within a table. Scenario 3 has gaps of 1,000 ns and 9,000 ns. Codes 42 and 43 are logging-only tests; they do not drive camera or LED outputs. Scenario 5 changes the event-101 rate, not the DG645 settings.

Results

All numbered scenarios

Scenario	Observed	Result
1	1 / 1 and 1 / 1 events received	Receiver difference 0 ns
2	2 / 2 and 2 / 2 events received	Receiver difference 0 ns; maximum gap error 0 ns
3	3 / 3 and 3 / 3 events received	Receiver difference 0 ns; maximum gap error 0 ns
4	40 / 40 and 40 / 40 events received	Receiver difference 0 ns; maximum gap error 0 ns
5	60 camera timestamps; 60 images	0 frame-ID gaps; 0 save overruns; LED-flash capture not established

`Machine-readable results.`

The differences below are reported timestamp differences at 8 ns resolution. They are not measurements of electrical output edges or source-to-receiver latency.

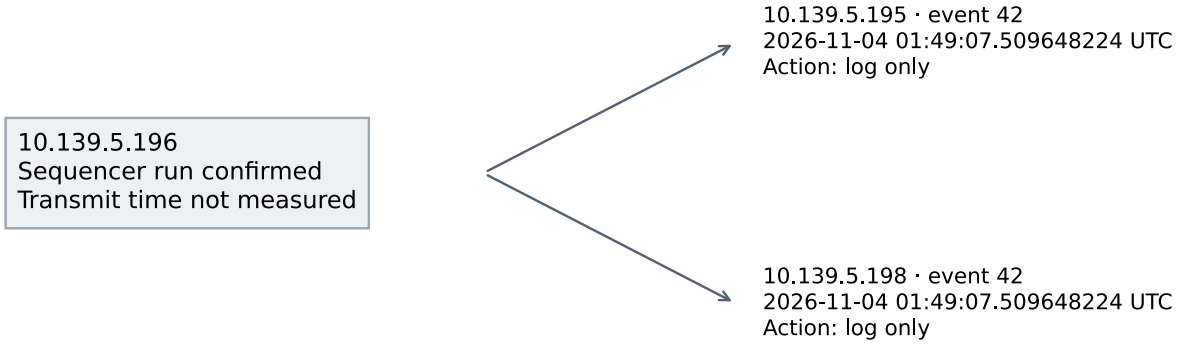
1. Single event

Programmed and received times

Event	Programmed (ns)	10.139.5.195 (ns)	10.139.5.198 (ns)	B − A (ns)
42	0	0	0	0

Reception timestamps

Event	10.139.5.195 — UTC	10.139.5.198 — UTC
42	2026-11-04 01:49:07.509648224 UTC	2026-11-04 01:49:07.509648224 UTC



Logical event path. Arrow length does not represent cable length or measured latency.

Recorded data: `scenario 1 event log`. Reference recording: `20261104T014907Z-4f26f2`.

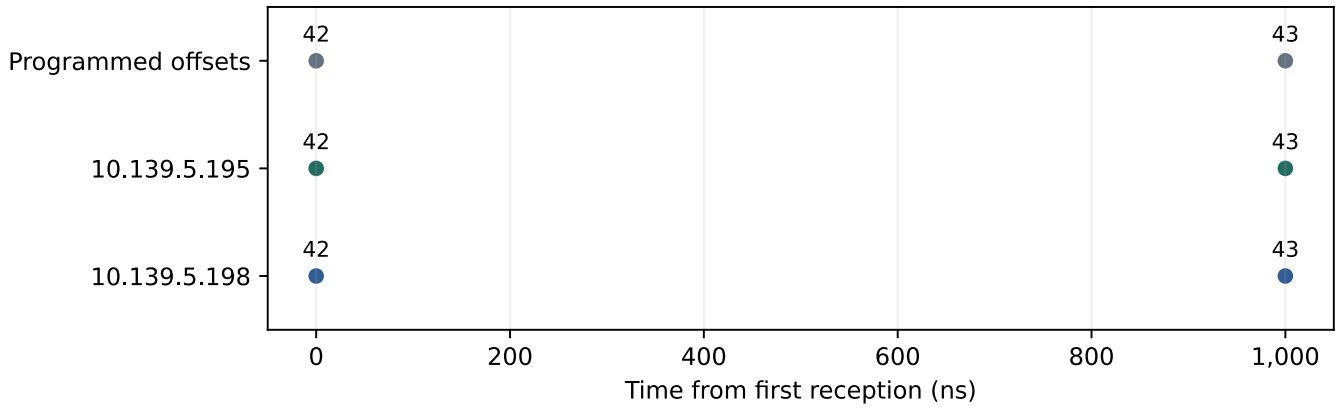
2. Two-event gap

Programmed and received times

Event	Programmed (ns)	10.139.5.195 (ns)	10.139.5.198 (ns)	B − A (ns)
42	0	0	0	0
43	1,000	1,000	1,000	0

Reception timestamps

Event	10.139.5.195 — UTC	10.139.5.198 — UTC
42	2026-11-04 00:28:34.494070584 UTC	2026-11-04 00:28:34.494070584 UTC
43	2026-11-04 00:28:34.494071584 UTC	2026-11-04 00:28:34.494071584 UTC



Numbers identify event codes. Each receiver's first reception is the time origin; source transmit time is not measured.

Recorded data: `scenario 2 event log`. Reference recording: `20261104T002834Z-e2c3c2`.

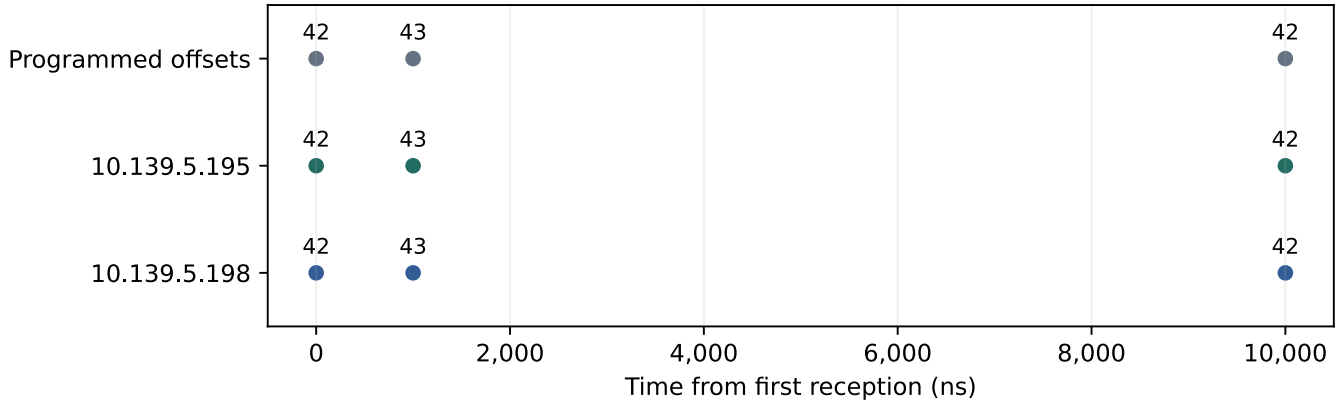
3. Multiple delays

Programmed and received times

Event	Programmed (ns)	10.139.5.195 (ns)	10.139.5.198 (ns)	B − A (ns)
42	0	0	0	0
43	1,000	1,000	1,000	0
42	10,000	10,000	10,000	0

Reception timestamps

Event	10.139.5.195 — UTC	10.139.5.198 — UTC
42	2026-11-04 01:49:23.410365384 UTC	2026-11-04 01:49:23.410365384 UTC
43	2026-11-04 01:49:23.410366384 UTC	2026-11-04 01:49:23.410366384 UTC
42	2026-11-04 01:49:23.410375384 UTC	2026-11-04 01:49:23.410375384 UTC



Numbers identify event codes. Each receiver's first reception is the time origin; source transmit time is not measured.

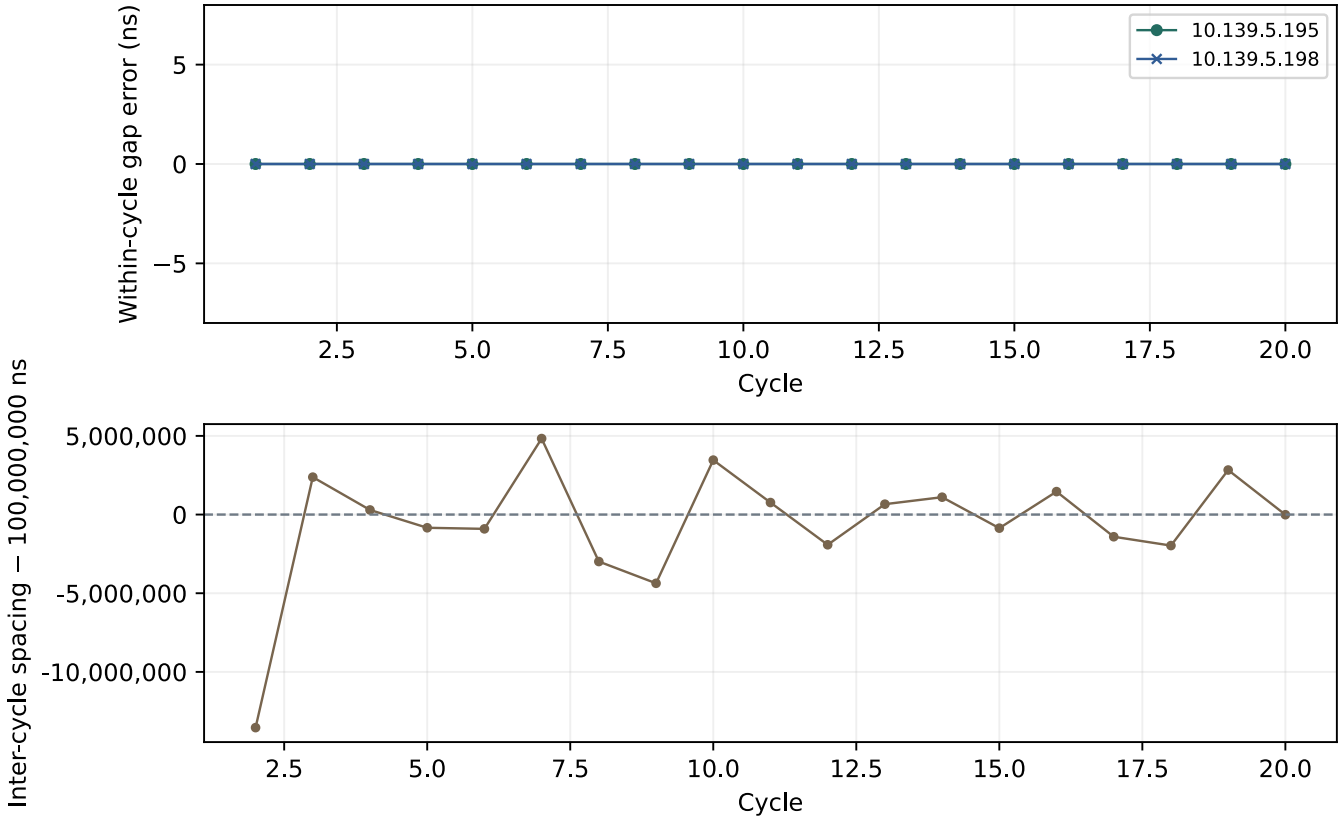
Recorded data: `scenario 3 event log`. Reference recording: `20261104T014923Z-d14cc7`.

4. Repeated Normal-mode sequence

Both receivers logged 40 of 40 events across 20 cycles. Every cycle retained the programmed 1,000 ns gap. The bank remained armed between software triggers. The final disarm and restoration were verified.

Within-cycle gap error

Receiver	Samples	Mean (ns)	σ (ns)	Minimum (ns)	Maximum (ns)
10.139.5.195	20	0	0	0	0
10.139.5.198	20	0	0	0	0



Top: FPGA-programmed gap stability. Bottom: received inter-cycle spacing under host scheduling. The lower trace does not measure oscillator jitter.

Received software-paced inter-cycle intervals

Mean (ns)	σ (ns)	Minimum (ns)	Maximum (ns)
99,419,162	3,771,209	86,459,256	104,831,104

The request cadence was nominally 100 ms. Its variation includes host scheduling, network communication and command verification. It is separate from the 1,000 ns gap generated inside each hardware table.

The hardware run counter wraps modulo 16. The verified counter sequence was `1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 0, 1, 2, 3, 4`. Twenty individual increments were checked. The final counter value of 4 alone would not demonstrate twenty runs. The four-bit width is defined in the tagged sequencer HDL.

▸ **Complete cycle and reception log**

Recorded data: `scenario 4 event log`. Reference recording: `20261104T022216Z-5c9f85`.

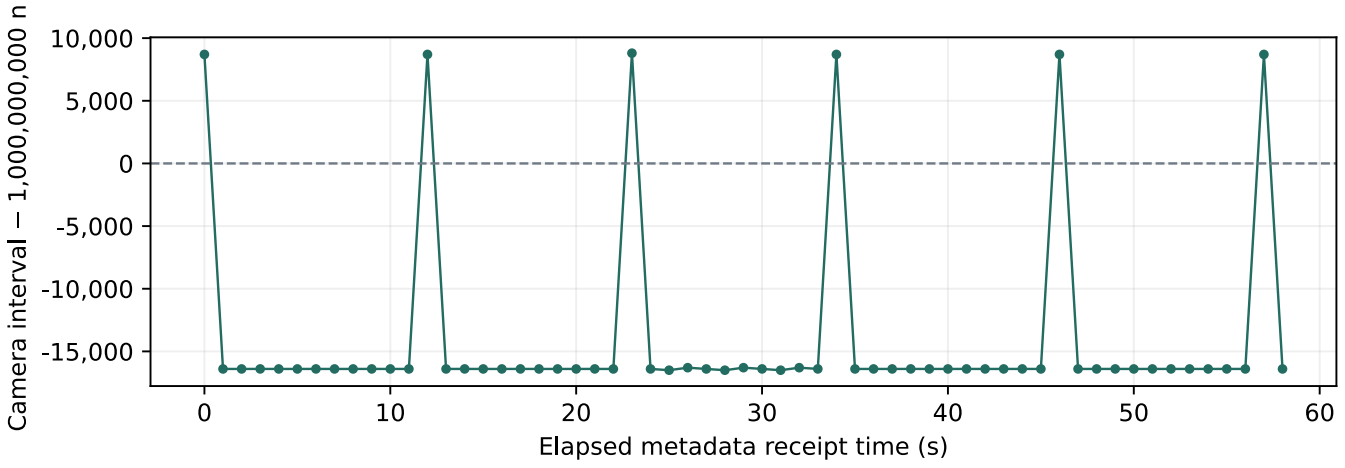
5. LED and camera response

At a configured event rate of 1 Hz, CAM2 supplied 60 timestamps and the recorder saved 60 full images. There were 0 frame-ID gaps, 0 writer overruns and 0 unassociated saved images.

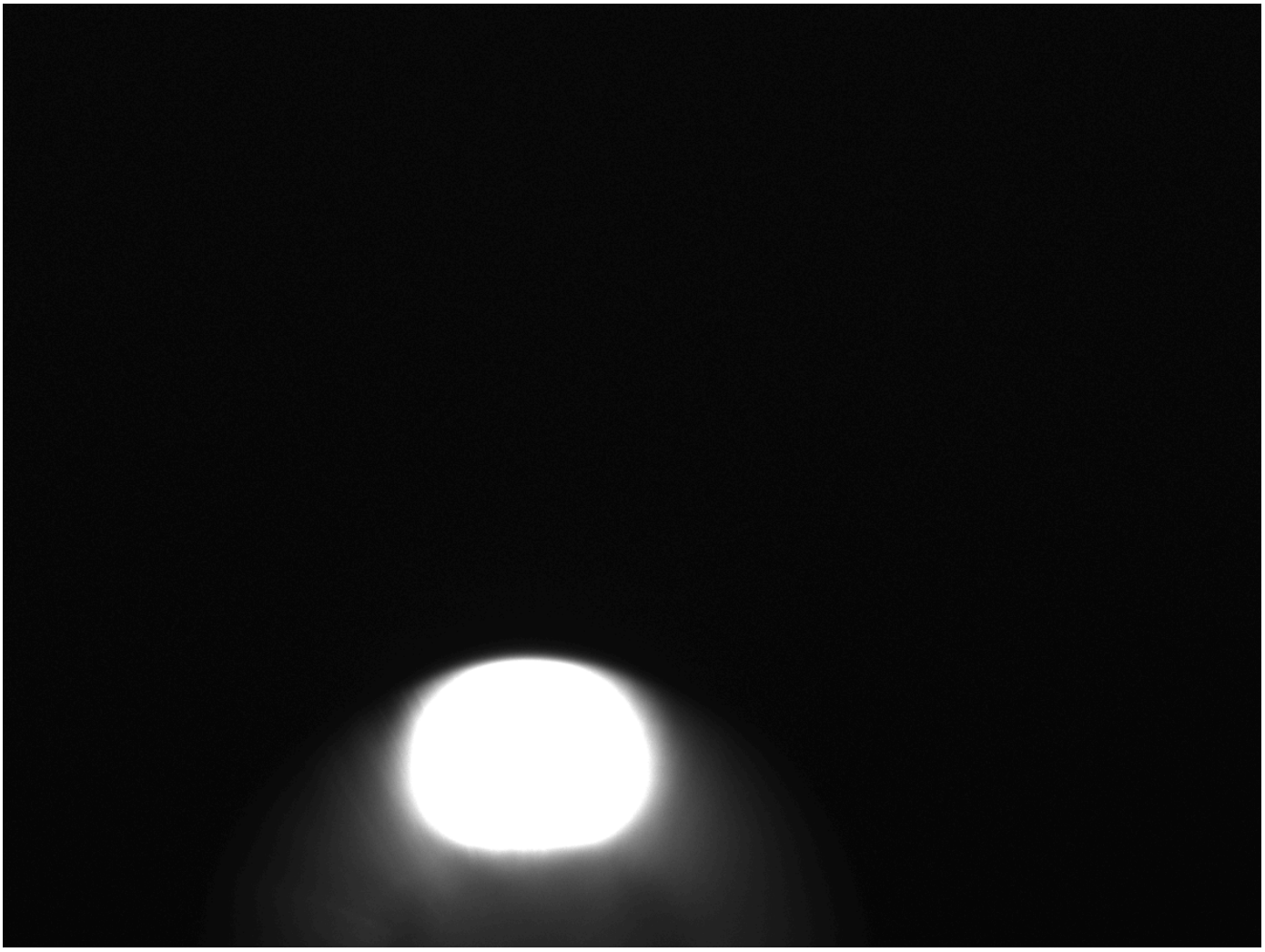
Camera delivery was demonstrated. Synchronized LED-flash capture was not. None of the 60 frames exceeded the fixed LED-bright threshold of 1000 counts in image columns 300–699 and rows 550–849. Region peaks were 172–191 counts. The data do not determine whether this was caused by phase, routing or the DG645 setup.

Measured response

Measurement	Result
Configured event-101 rate	1 Hz
Received camera rate	1.000014 frames/s
Mean camera interval	999,986,154 ns
Camera interval σ	7,592 ns
Compared receiver event pairs	58
Receiver difference mean / σ / min / max	0 / 0 / 0 / 0 ns



Camera timestamps measure frame spacing on the camera clock. They do not measure electrical trigger jitter.



10 Hz baseline: LED bright, frame 1343188.



1 Hz trial: LED dim, frame 1547826.

Both are display copies of saved 12-bit arrays. They are different test conditions; the bright reference is not presented as a successful 1 Hz exposure.

Camera clock resolution and its conversion to receiver time were not verified. A matching frame rate does not establish one identified timing event per frame. The 10 Hz reference shows the LED bright. The 1 Hz trial does not establish LED-flash capture. The data do not identify the connected DG645 output or the reason for the dim 1 Hz frames.

`Complete saved-frame metadata`. Reference recording: `20261104T013636Z-7f8d4f`.

Baseline and restoration

The unchanged 10 Hz baseline supplied 600 camera timestamps and saved 600 images. Its 580 compared receiver pairs had reported mean / σ / min / max differences of 0 / 0 / 0 / 0 ns.

Reference recordings

Scenario	Recording ID	Restoration
1	20261104T014907Z-4f26f2	Changed controls and full maps restored and verified
2	20261104T002834Z-e2c3c2	Changed controls and full maps restored and verified
3	20261104T014923Z-d14cc7	Changed controls and full maps restored and verified
4	20261104T022216Z-5c9f85	Changed controls and full maps restored and verified
5	20261104T013636Z-7f8d4f	Changed controls and full maps restored and verified

▸ **Exact applied, trigger and restoration commands**

▸ **Acquisition diagnostics and retained evidence**

Findings

All five numbered scenarios produced measured results. The compared receivers recorded the expected test codes and order. The finite patterns retained their programmed gaps. Their reported timestamp differences were 0 ns at 8 ns resolution.

Normal mode replayed the two-event table 20 times without rearming between cycles. All 40 events reached each compared receiver. Every within-cycle gap was 1,000 ns; the measured gap-error standard deviation was 0 ns at the available resolution. The bank was disarmed and idle before its original settings were restored.

These results support digital event delivery and reported timestamp consistency for CXFEL data correlation at the tested settings. They do not establish sub-tick skew, absolute UTC accuracy, cable delay or electrical trigger-to-exposure latency.

The camera followed the tested 10 Hz and 1 Hz rates without recorded frame-ID gaps or save overruns in the selected references. Camera frames still need verified event identifiers or a measured clock conversion before each exposure can be assigned to a facility timing event. None of the 60 frames in the 1 Hz trial showed an LED-bright state in the recorded region. Optical trigger alignment remains unresolved.

The 2,048-mark display limit does not limit trigger generation or the raw archive. The IOC reads at most 128 log entries per receiver per one-second scan, shared across logged codes. This configuration cannot sustain every shot at 1 kHz. The current 1 Gbit/s camera link also cannot carry 1,000 full-size images/s. Full-rate CXFEL operation remains unqualified.